BioTrace:

A Multimodal System for Real-Time Measurement of Stress, Cognitive Load, and Performance in Laporoscopic Training.

Group 3

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Abstract

Surgical doctors and surgical doctors in training often rely on external feedback when it comes to assessing their performance. Simulators like box trainers deliver objective feedback but are highly restricted to performance metrics, like time on task or accuracy. They lack physiological signals that provide real-time feedback and neglect physiological data such as stress or cognitive load. This can have a significant impact on surgical training, as it makes it difficult to understand the underlying causes of performance changes or surgical errors. Devices that combine these multimodal approaches do exist but are most often commercial and expensive. The goal of this work was to develop a cost-efficient and easily replicable measurement system that incorporates stress measurement via changes in HRV, cognitive workload via changes in pupil dilation, and performance metrics via accuracy and speed to provide insights about learning progress. Most importantly, all signals are bundled in a live dashboard to provide actionable insights. To simulate laparoscopy, participants performed box trainer exercises while our system detected all four signals synchronously after a calibration phase and visualized them in real time. All hardware designs, CAD models, and analysis scripts are available in the attached repository and are documented to a standard that enables reproduction. The system thus provides an accessible foundation for further research at the intersection of physiological measurement, cognitive workload, and medical simulation training, or even in other high-stress domains.

BioTrace

Surgical performance is critical because errors can have severe or fatal consequences. In laparoscopic surgery, performance is typically evaluated through speed and accuracy, reflecting the speed–accuracy trade-off (Heitz, 2014). However, performance is not solely determined by such technical skills. Research shows that stress and cognitive load significantly influence surgical performance outcomes (Tam et al., 2025; White et al., 2025). In fact, Laulan et al. (2025) found that 40% of surgeons have reported a direct link between intraoperative stress and surgical complications, as well as revealed that excessive workload or fatigue contributed to 33% of surgical errors.

In current laparoscopic training, only performance outcomes are visible, e.g., time taken or errors made. Surgeon's internal states of stress or mental workload during surgery remain unobservable in real time. This creates a gap: trainers cannot identify when performance breakdowns are caused by internal states. This lack of visibility in training extends to the operating room, where it becomes difficult to determine whether surgical errors were driven by underlying cognitive strain or stress, and/or external factors (Tam et al., 2025).

Researchers highlight the need for tools like heart-rate variability (HRV) and pupillometry to help trainers and institutions detect when a surgeon's cognitive capacity is being depleted before performance declines (Laulan et al., 2025). However, current research typically analyzes these signals in isolation and only after a task, missing real-time insights of surgeon's internal state during operations (Tam et al., 2025). Although combining these measures into a multimodal approach could improve stress detection, there is still a lack of integrated systems that provide real-time feedback. This points to a clear need for a platform that can link internal states to performance as they occur, enabling earlier and more informed interventions.

That said, this study proposes a multimodal system that makes internal states measurable by combining physiological and performance data in real time. The aim is to make otherwise invisible processes accessible to support more informed learning during training.

Research Question

Is it possible to build a multimodal system, a dashboard, that integrates HRV and pupil dilation data to reliably measure and visualize real-time stress and cognitive load during laparoscopic simulation training, and relate these measures to task performance?

Background

HRV as stress indicator

Stress refers to a physiological and psychological response to threatening or demanding situations. It is characterized by changes in the autonomic nervous system (ANS), particularly increased sympathetic activation. More specifically, it reflects a shift in the balance between the sympathetic and parasympathetic nervous system toward greater sympathetic activity, which leads to a decrease in physiological heart rate variability (Kim et al., 2018). Heart rate variability (HRV) captures this balance through the temporal variation of successive heartbeat intervals and is well established as a non-invasive marker for parasympathetic activity.

As our time-domain metric we use RMSSD, the root mean square of successive differences. We chose RMSSD because it directly reflects the vagally mediated HRV component, is robust to short recording windows, and is therefore well suited for real-time applications (Laborde et al., 2017; Shaffer & Ginsberg, 2017). Because RMSSD shows high inter-individual variance, we do not look at absolute values, rather at the change relative to an individual baseline recorded at the start of each session. A decrease in $\Delta\ln(\text{RMSSD})$ below -0.5 against the baseline is used as an indicator for elevated stress (Plews et al., 2013). For heartbeat detection we use the Pan-Tompkins algorithm, which is considered the de facto standard for real-time ECG processing due to its computational efficiency and has been validated on the MIT-BIH Arrhythmia Database with a detection rate of 99.3% (Pan & Tompkins, 1985).

Cognitive load

Cognitive load refers to the mental effort required to accomplish a task, which is directly related to working memory demands. It can be understood as an imbalance between the demands of a given task and the cognitive resources available at a given time, whereby high cognitive load can lead to performance degradation, fatigue, and similar effects (Abdurrahman et al., 2022). Physiologically, cognitive load is associated with sympathetic activation, which can be detected through pupil dilation, the so-called Task-Evoked Pupillary Response (TEPR) (Urrestilla & St-Onge, 2020). When cognitive demand decreases, the parasympathetic system becomes more dominant and the pupil constricts again (Jerčić et al., 2020).

Raw pupil size is not a reliable proxy for cognitive load, as it is simultaneously influenced by factors such as ambient light, accommodation, and emotional arousal (Urrestilla & St-Onge, 2020). We therefore measure cognitive load through the Percentage Change in Pupil Size (PCPS) relative to an individual baseline established during a calibration phase at the start of each session (Ma et al., 2024). During the session, a rolling live average of recent PCPS values is continuously calculated to account for gradual shifts caused by fatigue, adaptation, minor head movement, and environmental variation. This rolling average serves as the participant's current workload reference level. An adaptive threshold, the Workload Index Value (WIV), is then calculated as the live average multiplied by a threshold adapter, which acts as a sensitivity coefficient and is dynamically adjusted throughout the session: if too few meaningful changes are detected the sensitivity increases, and if too many fluctuations are detected it decreases. Elevated cognitive load is indicated when PCPS exceeds the WIV, while values below the WIV suggest lower workload or a recovery moment. Given the absence of universal thresholds in the literature, all results are interpreted relative to the individual baseline and task context (Duchowski et al., 2018).

Performance and learning curves

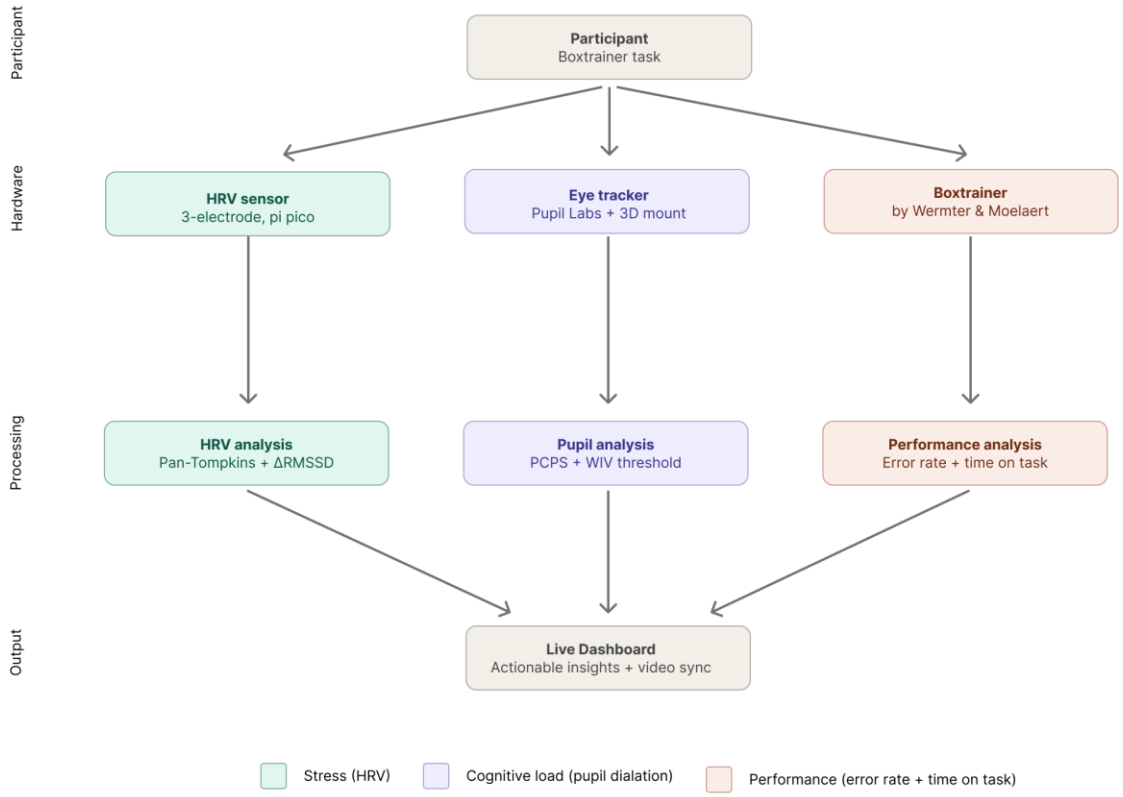
To measure the progress of participants, we record two performance metrics per session: wall contacts as a measure of accuracy and time on task as a measure of speed. Both values are expected to decrease over repeated training sessions. To model this progress, we use a parametric learning curve approach developed by Schmettow (github.com/schmettow/pub-learning-curves-surgery): $perf(t) = (1 - leff)^{(t+pexp)} + \max p$.

The model is built around three parameters: Previous Experience (pexp), which shifts the curve along the trial axis and captures how much prior experience someone brings in, Learning Efficiency (leff), which determines the curvature and therefore how quickly someone improves, and Maximum Performance (maxp), which represents the theoretical performance ceiling a person can reach. This allows us to not only track where a participant currently stands, but also to get insights into their individual learning speed and their personal performance potential.

Systemic overview

The system (BioTrace) integrates three hardware components: a custom-built ECG sensor using three electrodes and a Raspberry Pi Pico, a Pupil Labs Core eye tracker modified with a custom 3D-printed mount and a laparoscopic box trainer developed by Wermter and Moelaert. To ensure data accuracy across participants we are running a 60-second calibration phase at the start of each session to compute individual baseline RMSSD (ms) and baseline pupil diameter (px) values, which serve as reference points for all subsequent analysis. During the live session, the HRV pipeline applies Pan-Tompkins QRS detection to monitor delta-RMSSD changes relative to this baseline to detect stress events (Section 4.1). At the same time, the pupil pipeline tracks dilation changes using an adaptive WIV threshold to identify high cognitive workload events (Section 4.2). Performance is captured via wall contacts and task completion time per trial, which are used post-session for learning curve modelling following Schmettow (Section 4.3).

Figure 1. Full set-up overview.



All outputs are bundled in a live dashboard linked to a video recording of the session. The three signal streams operate at different sampling rates: raw ECG at 150 Hz, RMSSD computed every 5 seconds over a 30-second sliding window (0.2 Hz), raw pupil diameter at approximately 30 Hz, and PDI and CLI at up to 30 Hz after artifact rejection. All streams are synchronized via a shared session timer and normalised to a common zero point at session start (elapsed = timestamp - self._session_start_time).

Methods

Eye-tracker

Apparatus

The eye-tracking set-up consisted of a USB endoscope camera with 5.5 mm diameter, an infrared (IR) light, and the Thonny application for programming the software. To power the IR

light, a battery holder; three AA batteries; and wires to power the IR light were used (see figure 2).

An arm and camera holder were built that allowed flexible position adjustment of the eye-tracker. To do this the following materials were used: three 3D printed vertical links which allows vertical movement (Vertical_link.stl); three 3D printed horizontal links which allows horizontal movement (Horizontal_link.stl); six M3 nuts and bolts to connect the links; a 3D printed camera holder (Camera_attachment.stl) and a 3D printed attachment (Frame_attachment.stl) that attaches the camera holder to the main frame.

The main frame of the eye-tracker was a 3D printed frame which was adopted from Pupil Labs' open GitHub repository found in (Kassner, 2026). All unnecessary attachments were discarded of, the final frame consisted of the side triangle and the main frame (see figure 4). Please refer to the GitHub repository for the open source STL files.

Method of Assembly

As the first step of the set-up, the IR light was powered using the materials in Figure 2 along with three AA batteries. It is important to note that the power source of the IR light cannot exceed five volts to prevent the light from burning. The cables of the battery holder were manually soldered to the IR light to establish a power source for it. Once soldered, the batteries were inserted to confirm the functioning of the IR light.

Figure 2. Materials used for the eye-tracker.

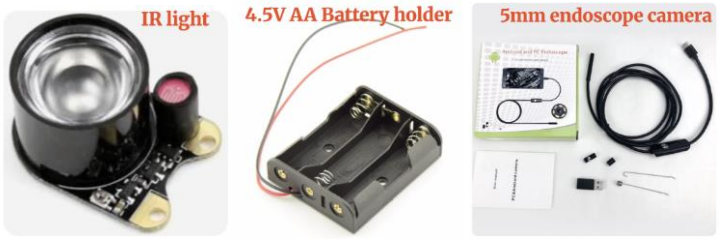
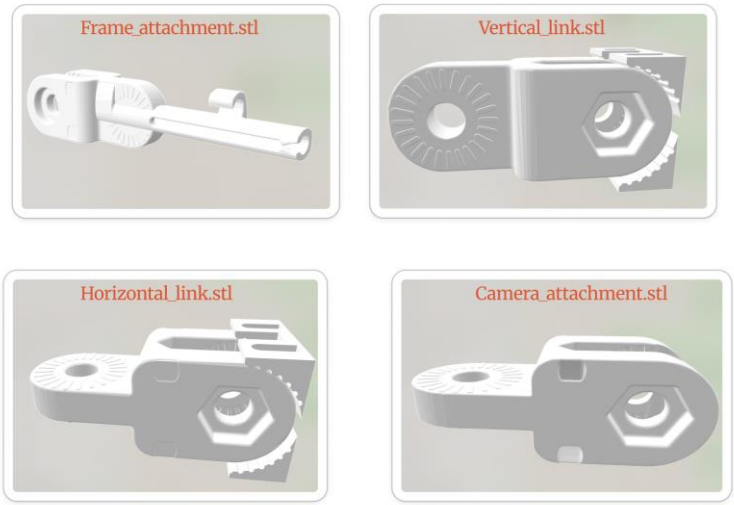
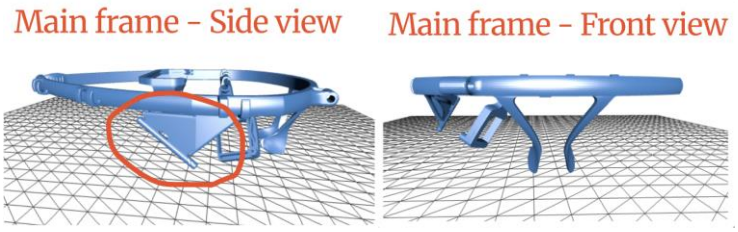


Figure 3. Attachments for the arm and camera holder.



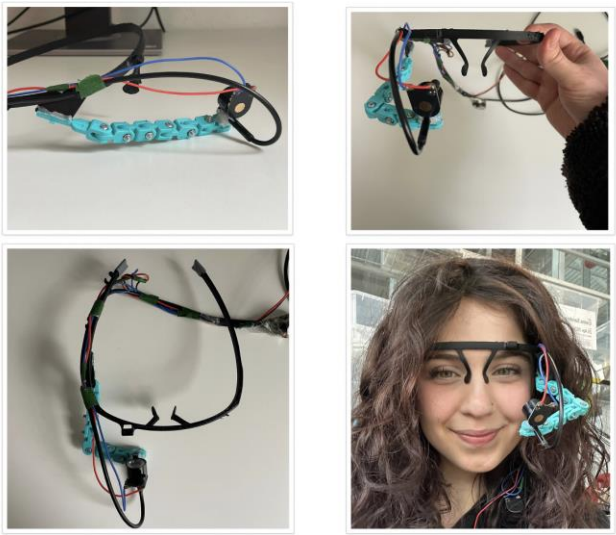
To build the full frame for the eye-tracker, the required 3D printable attachments (figure 3) and the mound (figure 4) were printed before proceeding with the assembly. First, the arm of the full frame was assembled. This involved starting with the frame attachment and subsequently connecting it to a vertical attachment using an M3 nut and bolt. The arm was then extended through alternating horizontal and vertical links, ending with the camera attachment (see Figure 3). The long end of the frame attachment was connected to the triangular part of the main frame (see highlighted section in figure 4). Hot glue was used to attach the camera to the other end of the arm. Lastly, the IR light was glued on top of the camera with the round side facing down. This ensured that the camera and IR light remained at approximately the same distance from the eye and moved in synchrony. Figure 5 presents the final set up.

Figure 4. The main frame (Mound)[1]



[1]*Sourced from https://github.com/pupil-labs/pupil-geometry*

Figure 5. The final set-up



Pupil Diameter and Cognitive Load

To measure pupil dynamics, a USB eye-tracking system was used which was integrated into the final dashboard environment. For an overview of the entire process, refer to figure 7. The camera was continuously capturing video frames of the eye in real time. These frames were converted to grayscale prior to measurement to improve contrast between the iris and the pupil. The PuRe pupil detection algorithm (Santini et al., 2018), was used to process each video frame. This algorithm has been designed for real-time pupil detection under non-ideal imaging conditions, for example one where continuous movement is involved like laparoscopy. PuRe identifies dark, circular or elliptical contours consistent with expected pupil geometry while rejecting artefacts caused by eye lids, eyelashes, reflections or image noise (Santini et al., 2018).

To improve tracking stability, the system initially performed detection within a pre-determined region of interest (ROI); in case of an invalid detection, the algorithm reverted to a full frame search. Furthermore, to reduce frame to frame jitter while preserving real time responsiveness, an exponential moving average (EMA) filter was applied. To apply this a smoothing coefficient of $\alpha = 0.30$ was used as a moderate weighting parameter (Vishwas & Sumana, 2026).

Pupil diameter was measured in pixels for each frame and time-stamped for synchronization with the remaining physiological signals. From these raw pupil diameter values, percentages of change in pupil size (PCPS) were measured relative to an individual resting baseline collected at the beginning of each session (see formula in figure 6). Baseline normalization was used to enable comparison across participants or sessions.

Figure 6. PCPS formula.

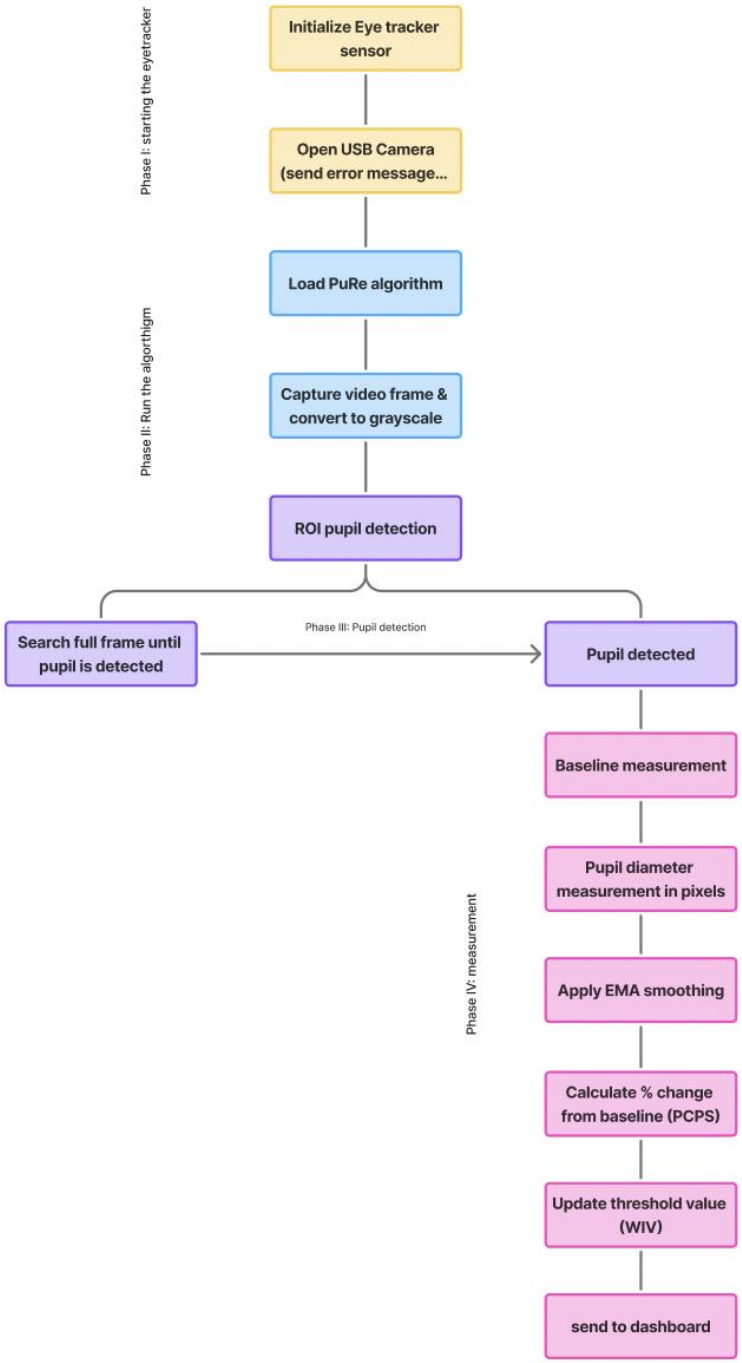
$$\%\Delta\text{Pupil} = \frac{Current - Baseline}{Baseline} \times 100$$

Cognitive load

High cognitive load events were identified using an adaptive thresholding approach rather than a fixed pupil-size cutoff. The Workload Indicator Value (WIV), adapted from Katidioti et al. (2016), was implemented as the smoothed Pupil Dilation Index (PDI), with a 1-second smoothing window applied to decrease short-term noise. A rolling 30-second mean of the smoothed signal was then calculated to provide a continuously updating reference level based on the participant's recent pupil behaviour.

The decision threshold was defined as *threshold = rolling mean × 0.997*, placing the cutoff slightly below the recent average signal. This multiplier was originally selected through pilot testing by Katidoti et al. (2016), to improve sensitivity to workload-related changes while avoiding excessive instability. Considering the scope of the project, we did not perform an independent pilot test to identify a different and instead adopted the same value. The aim of the multiplier (*0.997*) is to create a buffer that makes the system more stable and sensitive to increases above normal recent levels. Cognitive load was classified as high when the smoothed PCPS exceeded the adaptive threshold and low otherwise (Katidoti et al., 2016).

Figure 7. Process overview of pupil measurement


Phase I: starting the eyetracker
Initialize Eye tracker sensor
Open USB Camera (send error message...
Phase II: Run the algorthigm
Load PuRe algorithm
Capture video frame & convert to grayscale
ROI pupil detection
Search full frame until pupil is detected
Phase III: Pupil detection
Pupil detected
Phase IV: measurement
Baseline measurement
Pupil diameter measurement in pixels
Apply EMA smoothing
Calculate % change from baseline (PCPS)
Update threshold value (WIV)
send to dashboard

HRV Sensor

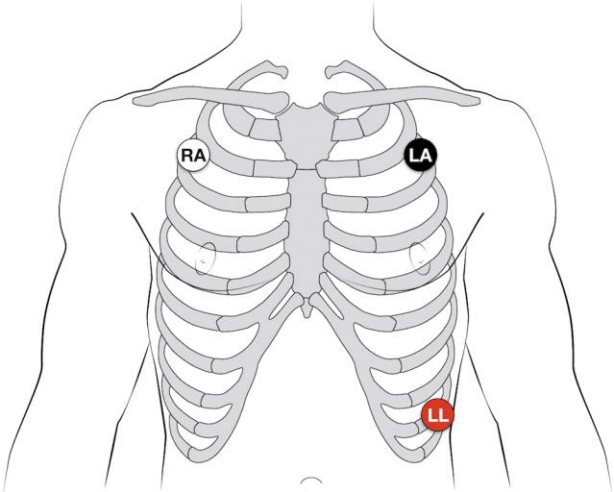
Apparatus

To build the HRV sensor the following components were used: three ECG electrodes; one AD8232 ECG signal amplifier module; a Raspberry Pi Pico microcontroller; connection cables between the electrodes and the AD8232; as well as connection cables between the AD8232 and the Pico; and a USB cable to connect the Pico to the host computer. On the Raspberry Pi Pico YLab (github.com/schmettow/YLab) was installed.

Assembly

The first part of the set-up involved attaching the electrodes to the body as shown in figure 8. The electrodes were connected to the AD8232, which amplified the weak electrical heart signal before transferring it to the Pico. The output of the AD8232 was connected to pin GP26 of the Raspberry Pi Pico. On the Pico the Sensory.py and code.py files from YLab were used. Once everything was connected, GP20 was pressed and once the LED light turned green, it was confirmed that the system is recording. At this point the Pico started sending data at 150 Hz to the connected computer.

Figure 8. Electrode placement.

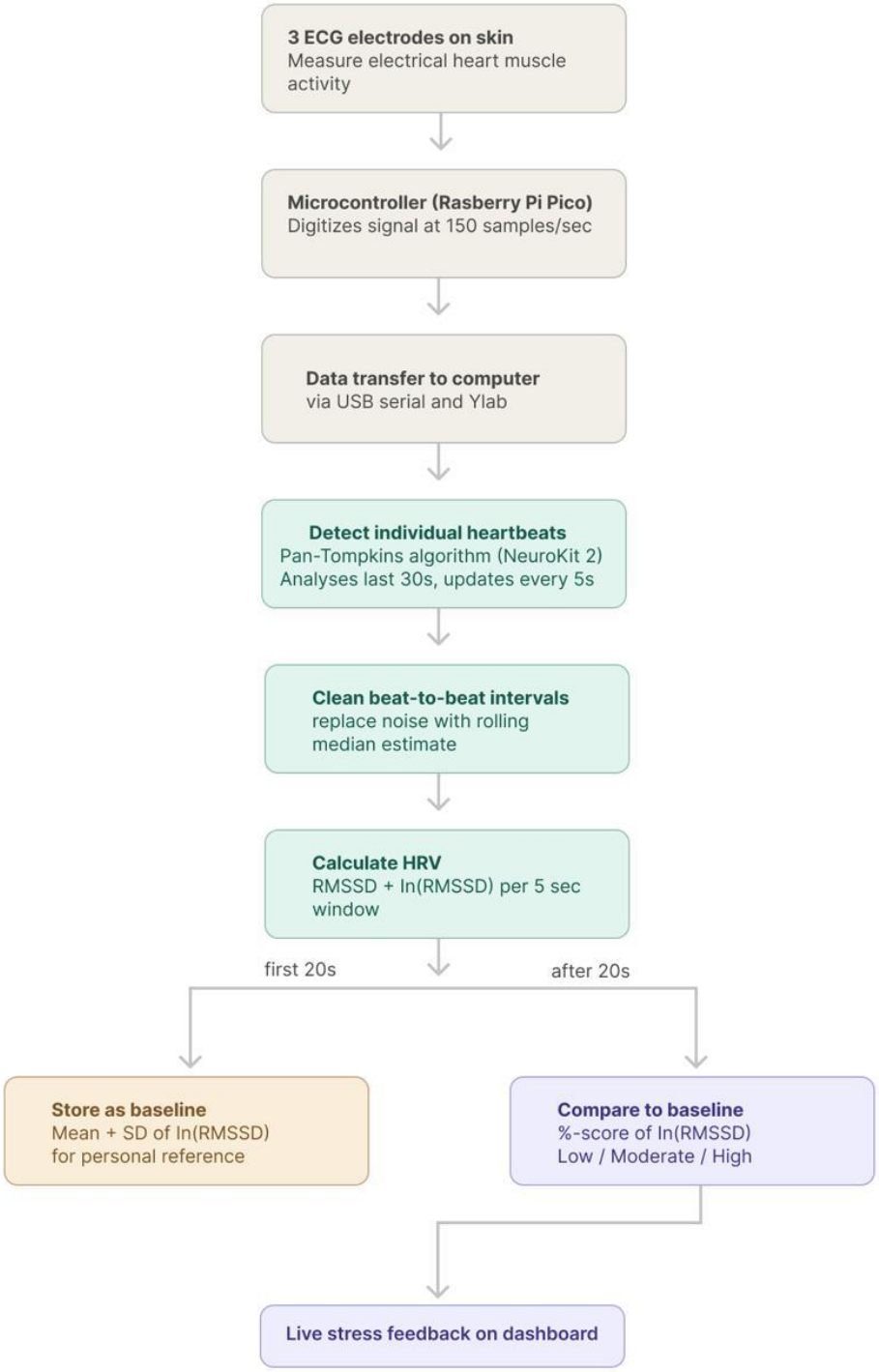


Software and signal processing

The software pipeline converted the raw electrode signal into a live stress value in the dashboard. The complete processing pipeline is summarized in Figure 9 and consists of four phases. In Phase I, the three electrodes captured the electrical signal of the heart muscle. The AD8232 amplified this signal before it was digitized by the Pico at 150 Hz and sent to the host

computer via USB. In Phase II the Pan-Tompkins algorithm was used and implemented via the NeuroKit2 library, to detect individual heartbeats in the electrical signal. The analysis ran on a 30-second sliding window that was updated every 5 seconds. In Phase III, the intervals between consecutive heartbeats were cleaned. Values outside the physiologically plausible range of 300 to 2000 ms are marked as invalid and replaced using a rolling median estimate. From the cleaned intervals, the RMSSD value and its natural logarithm were then calculated. In Phase IV, the current RMSSD value was compared to the individual baseline recorded at the start of each session. The delta-RMSSD was calculated, which is the relative change between the two values. A negative delta-RMSSD indicates increased sympathetic activation, which we interpret as a stress response. To determine the extent of stress we use the following thresholds: Stable: 0 to -10% Mild stress: -10% to -40% High stress: below -40% This stress value is then displayed live in the dashboard.

Figure 9. Process overview of HRV measurements.



Box trainer

Apparatus and Assembly

The box trainer used for the demonstration was built by Wermter and Moelaert (n.d.). For a detailed description of the necessary materials and instructions on how to build the box trainer refer to their report attached in the BioTrace_V0 GitHub repository. Alongside the box trainer, two laparoscopic instruments, a Raspberry Pi Pico, a USB endoscope camera with 5.5 mm diameter, and two H7 Cables with female pin and clip are required for this set up.

To assemble the box trainer, one of the clips was attached to the mesh wire inside the box trainer, with its corresponding pin connected to the ground pin on the PiPico board. The other clip was placed on the right instrument with its corresponding female pin connected to GP1 on the PiPico board. The PiPico was then connected to a laptop to allow signal processing. The camera was taped to the left instrument to allow real time recording of the procedure. This was powered using a laptop.

Wall Contact detection

Wall contact was measured using contact sensors connected to the Raspberry Pi Pico and streamed through the same USB serial connection as the ECG data. During the laparoscopic box-trainer task, when the surgical instrument touched the wall or wire boundary, the sensor changed from a low state (0) to a high state (1). The software continuously monitored these signals (MOI channels) and counted each low-to-high transition as one wall-contact error. To avoid multiple counts from a single prolonged touch or signal bounce, a minimum time interval was applied between detections. Each valid contact event was then sent to the dashboard in real time, where it was added to the total error count. The full code can be found in

Dashboard software

BioTrace was developed as a real-time system that combined several hardware data streams into a single live dashboard. ECG data, pupil measurements, and wall-contact signals were collected continuously throughout each session. All incoming data were time-stamped and synchronized to a shared session timeline so that changes across measures could be viewed together in real time.

A central session management layer was used to control the flow of data between the hardware sensors, processing modules, and dashboard interfaces. Signals received from each

device were routed to their respective processing components, where raw measurements were converted into meaningful performance indicators. ECG data were transformed into RR intervals and used to calculate RMSSD as the heart rate variability metric. Pupil diameter values were converted into percentage change scores relative to the calibration baseline to indicate changes in cognitive load. Wall-contact signals were detected as discrete error events and counted individually.

The processed outputs were then sent directly to the live dashboard for continuous visual updates. RMSSD values were used to update the stress indicator and HRV timeline graph. Pupil metrics were used to update the workload indicators and threshold-based cognitive load state. Error events were used to update the live error counter whenever contact occurred. Because all signals were linked to the same session clock, the interface allowed physiological stress, cognitive load, pupil behaviour, and task-performance errors to be displayed simultaneously in one synchronized dashboard. This enabled users to monitor multiple aspects of performance at the same time during training sessions. Pupil-derived signals were used to drive workload displays and classify high cognitive-load events. Error-event signals updated the live error counter whenever wall-contact occurred. A presentation of the final live dashboard is shown in figure 10

Figure 10. BioTrace monitoring stress, cognitive load, and error rate in real time

Discussion

This study aimed to explore whether a multimodal system integrating HRV, pupil dilation, and performance metrics can make internal states like stress and cognitive load measurable and interpretable in real time during laparoscopic training. BioTrace demonstrates that it is feasible to capture, synchronize, and visualize these signals within a single dashboard, in a cost-efficient manner.

A key contribution of this study lies in translating physiological signals into actionable insights. By normalizing HRV and pupil dilation relative to individual baselines and presenting them alongside performance metrics, the system allows for meaningful interpretation of internal states during a task. This aligns with prior research suggesting that multimodal approaches can improve robustness in dynamic and high-risk environments, such as surgery. The real-time nature of the dashboard enables immediate awareness of differences in stress and cognitive load, which is not possible in traditional post-hoc analyses.

However, there are some limitations that must be taken into account. Firstly, the system relies on indirect physiological proxies (RMSSD and pupil dilation) that are sensitive to external factors like movement, lighting conditions, and individual variability. Although baseline

normalization and adaptive thresholds were implemented to mitigate these effects, the lack of a universal cut-off or validation limits our ability to fully confirm the accuracy of the inferred states. Secondly, the hardware setup, the wired ECG and DIY eye-tracker, may limit usability, general accuracy, and make the system less representative of real surgical conditions. Furthermore, due to the simple setup of the box trainer, we were limited in the amount and depth of information regarding performance. Lastly, while the dashboard integrates multiple signals, the current implementation treats stress and cognitive load as separate indicators rather than a unified index.

Future work should therefore focus on improving both validity and usability. From a measurement perspective, integrating additional modalities such as EEG or validated subjective scales (e.g., NASA-TLX) would strengthen the interpretation of cognitive load and stress. Additionally, it would be valuable to run some usability tests with surgical trainees as well. Moreover, developing a composite cognitive load index that combines HRV and pupillometry could provide a more holistic representation of internal states. More research needs to be dedicated to developing such a measurement. Furthermore, enhancing the hardware setup through wireless integration would improve user comfort and applicability in real-world training environments.

To conclude, this study provides proof of concept for making internal physiological states visible during laparoscopic training. By linking stress, cognitive load, and performance in real time, the system offers a foundation for more informed feedback and adaptive training approaches. This has the potential to support not only skill acquisition but also safer performance in high-risk contexts.

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